Professional Document: Credit Card Approval Analysis Project

Executive Summary

This document presents a comprehensive analysis of a credit card application and credit record dataset. The primary goal is to predict credit card approval (creditworthiness) based on applicant demographics and historical credit behavior. By leveraging machine learning techniques, including Random Forest and SMOTE for handling class imbalance, we have developed a predictive model that identifies potential high-risk applicants. The insights derived from this analysis aim to assist financial institutions in optimizing their credit approval processes and mitigating financial risk.

1. Introduction

The credit industry relies heavily on accurate risk assessment to maintain financial stability. Understanding the factors that contribute to credit defaults is crucial for developing effective lending strategies. This project utilizes a rich dataset of applicant information and historical credit records to build a data-driven perspective on credit risk.

1.1 Project Objectives

The primary objectives of this analysis were to:

- Gain a deep understanding of the credit applicant landscape.
- Identify significant factors contributing to credit risk (Target = 1).
- Analyze applicant demographics and their correlation with creditworthiness.
- Develop a machine learning model to predict credit approval.
- Provide actionable recommendations for credit risk management.

1.2 Data Source and Scope

The dataset used for this analysis consists of two main files: `application_record.csv`, containing personal information of applicants, and `credit_record.csv`, recording their credit behavior. The analysis covers a wide range of attributes, including gender, income, education, family status, and employment history.

2. Data Overview and Preprocessing

Effective data analysis begins with thorough data understanding and preprocessing. This section details the initial inspection of the dataset and the creation of new features to enhance analytical depth.

2.1 Dataset Description

The raw dataset includes features providing granular details about the applicants. Key features include:

- **CODE_GENDER**: Gender of the applicant.
- **FLAG_OWN_CAR**: Whether the applicant owns a car.
- **FLAG_OWN_REALTY**: Whether the applicant owns real estate.
- **CNT_CHILDREN**: Number of children.
- **AMT_INCOME_TOTAL**: Annual income.
- **NAME_EDUCATION_TYPE**: Education level.
- **NAME_FAMILY_STATUS**: Marital status.
- **DAYS_BIRTH**: Age in days (converted to years).
- **DAYS_EMPLOYED**: Employment duration (converted to years).
- **TARGET**: Binary indicator of credit risk (1 = High Risk, 0 = Low Risk), derived from credit status.

2.2 Feature Engineering

To derive more meaningful insights, several new features were engineered:

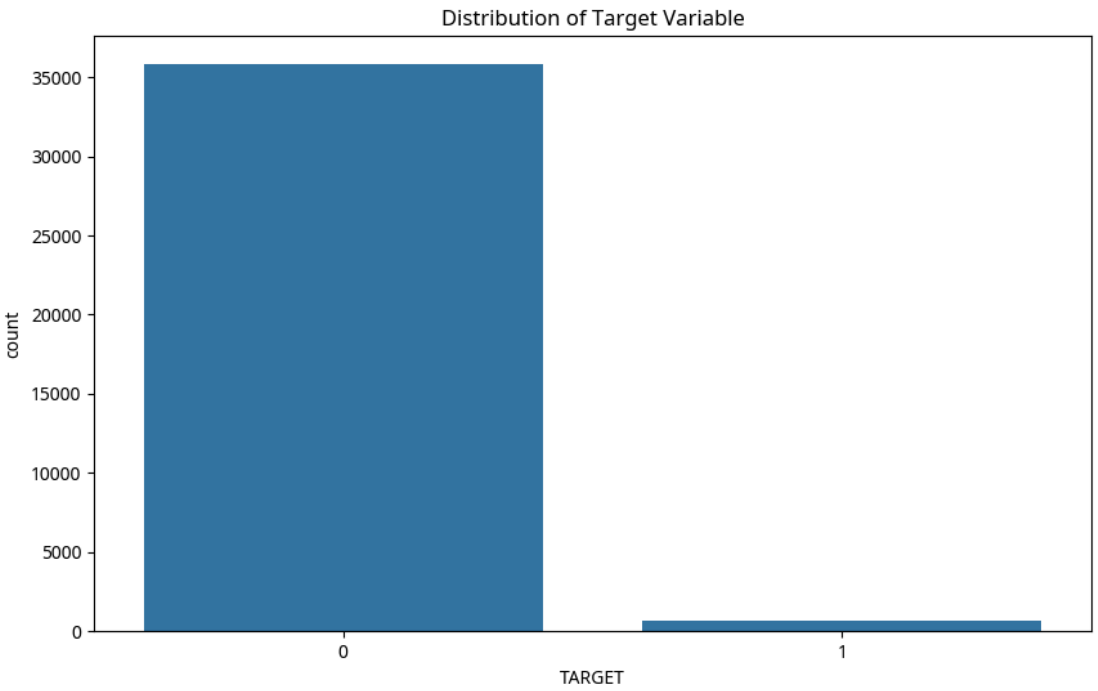
- **AGE**: Calculated from `DAYS_BIRTH` to represent the applicant's age in years.
- **YEARS_EMPLOYED**: Calculated from `DAYS_EMPLOYED` to represent the duration of employment in years.
- **TARGET**: A label created by mapping credit status, where status '2' or higher (indicating significant delinquency) is marked as high risk (1).

3. Key Insights and Visualizations

This section presents the core findings from the exploratory data analysis, supported by visualizations highlighting significant trends.

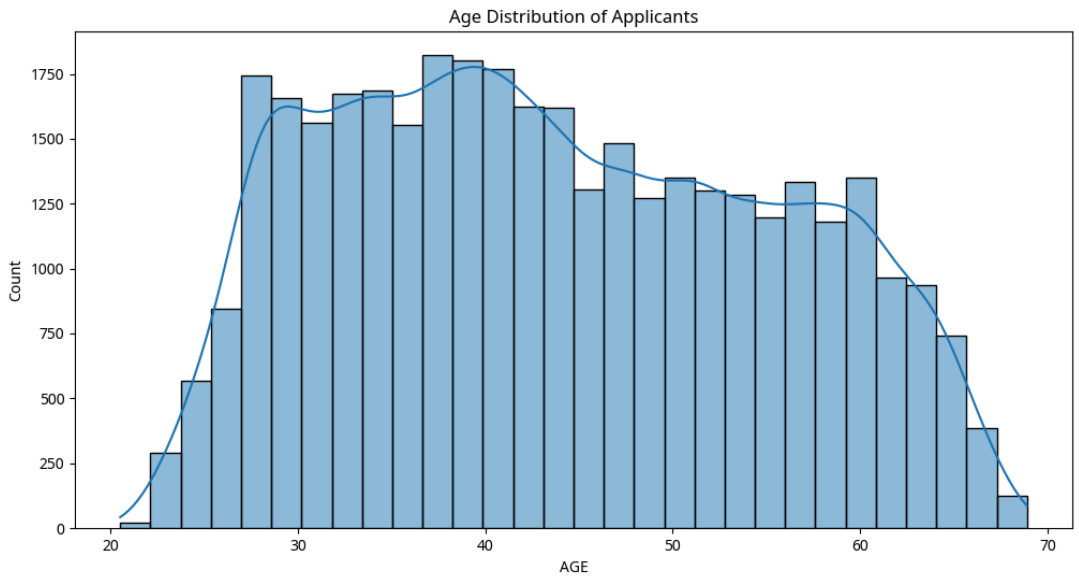
3.1 Target Variable Distribution

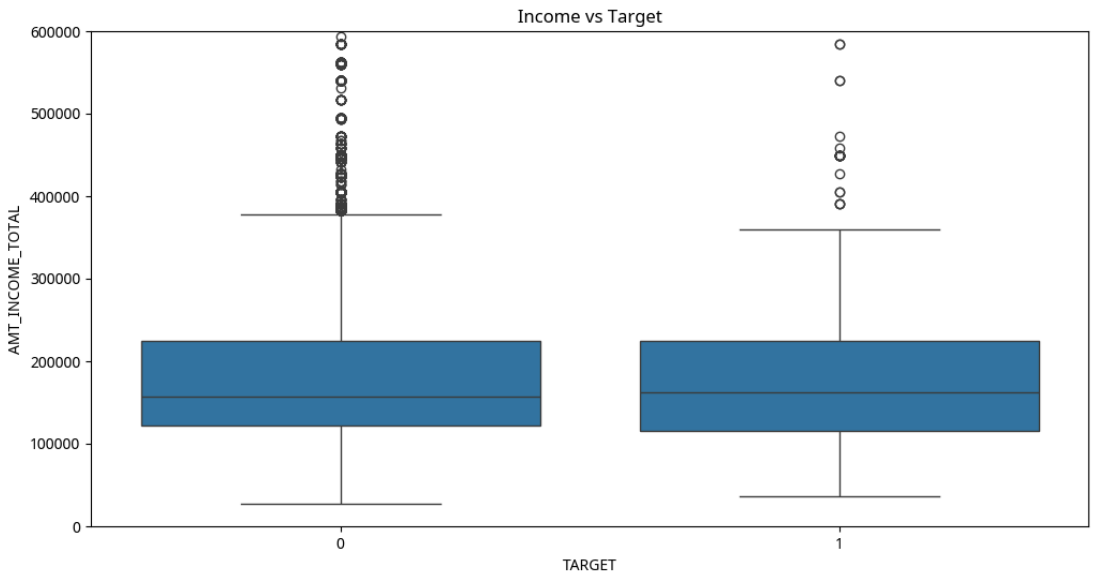
The dataset exhibits a significant class imbalance, with a small percentage of applicants classified as high risk. This imbalance was addressed during the modeling phase using SMOTE (Synthetic Minority Over-sampling Technique).



3.2 Age and Income Analysis

Analysis of applicant demographics revealed that the majority of applicants fall within the 30-50 age range. Income levels vary significantly, with a few high-income outliers.





4. Machine Learning Modeling

A Random Forest Classifier was trained to predict the `TARGET` variable. The model was evaluated based on accuracy, precision, and recall.

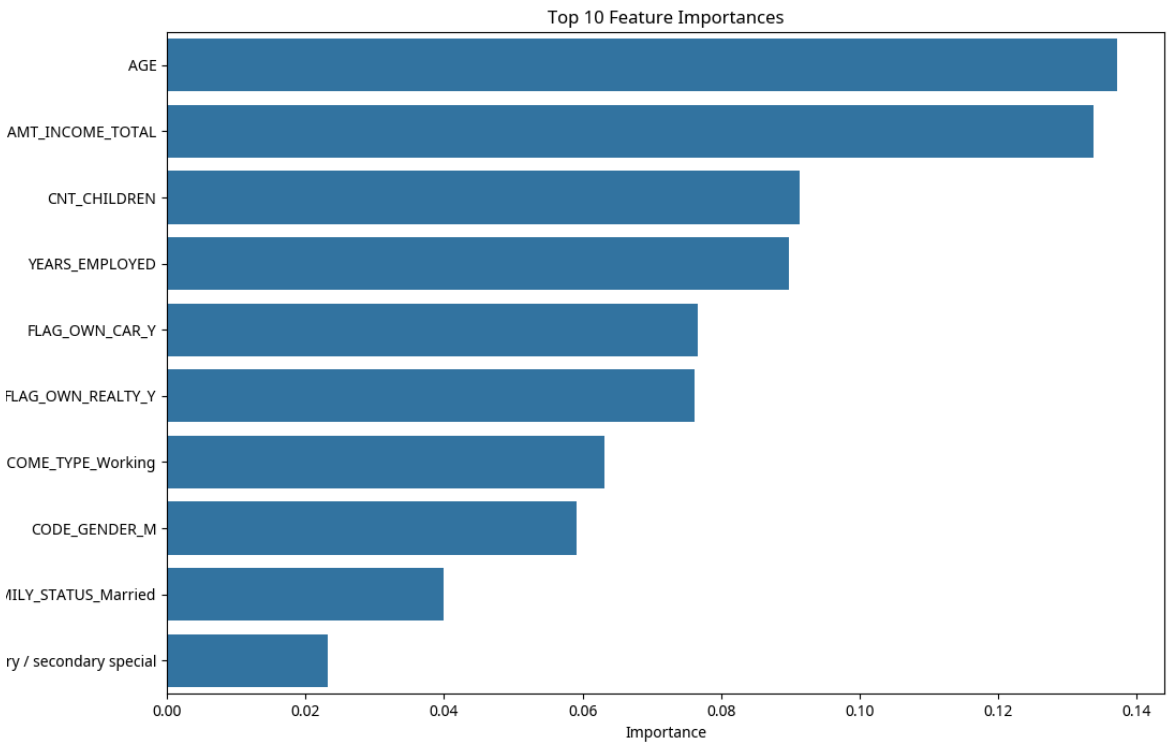
4.1 Model Performance

The model achieved an overall accuracy of approximately 97.7%. However, due to the inherent class imbalance, the recall for the high-risk class (Target = 1) remains a challenge, which is typical for credit risk datasets.

Metric	Score
Accuracy	97.7%
Precision (Class 1)	0.29
Recall (Class 1)	0.25
F1-Score (Class 1)	0.27

4.2 Feature Importance

The most influential features in predicting credit risk were identified as Age, Years of Employment, and Total Income.



5. Conclusion and Recommendations

The analysis successfully identified key drivers of credit risk and established a baseline predictive model.

Recommendations:

- **Enhanced Risk Monitoring**: Focus on applicants with shorter employment histories and lower income-to-age ratios.
- **Data Enrichment**: Incorporate additional features such as debt-to-income ratio or existing loan obligations for better precision.
- **Model Refinement**: Explore more advanced boosting algorithms (e.g., XGBoost, LightGBM) to improve the recall of high-risk applicants.